

Scalability & Future Plan

Date	5 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan

S.No	Limitation	Impact	Priority to Address
1.	Single-user local JSON storage	Cannot support concurrent multi-user access or persistence across devices	Medium
2.	GPT-2 Small generation quality	Occasionally produces short or generic conversation starters	Medium
3.	Sequential CPU inference, no caching	10-20 second response time per generation	Low — acceptable for current single-user demo scope

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1.	User Load	Single local user only	Add a lightweight database (SQLite/PostgreSQL) and basic session handling to support multiple concurrent users
2.	Data Storage	Local JSON files	Migrate to a relational or document database for structured, queryable, multi-user storage
3.	Performance	Sequential CPU inference, no caching	Cache repeated Wikipedia lookups; consider GPU inference or a hosted inference API for faster generation
4.	Security	No authentication implemented	Add basic authentication if the app is ever exposed beyond local single-user use

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Persistent user profiles	Short-term (next iteration)	Enables cross-session personalization
Phase 2	Fine-tune GPT-2 or upgrade to a larger instruction-tuned model	Medium-term	Improves conversation starter quality, reduces genericness
Phase 3	Multi-language support	Medium-term	Expands usability to non-English-speaking events
Phase 4	Live AI chat companion + attendee-matching algorithm	Long-term	Delivers the two high-effort/high-impact ideas deliberately deferred in Folder 1's Brainstorming & Idea Prioritization